PROJECT + SETUP + DAILY OPERATIONS

AI customer support + lead inquiry CRM

How inquiries become leads, how AI replies are reviewed, where your data lives, and how the dashboard connects to n8n, Gmail and Google Sheets.

Capture
Form or email

→

Understand
AI + lead record

→

Review
Owner dashboard

→

Act
Sync or confirm send

Your current setup

Business Gmail: adamromas0634@gmail.com

Dashboard: http://localhost:3001

Automation: Docker n8n at http://localhost:5678

Live workbook: AI-SUPPORT CRM WORKFLOW in Google Sheets

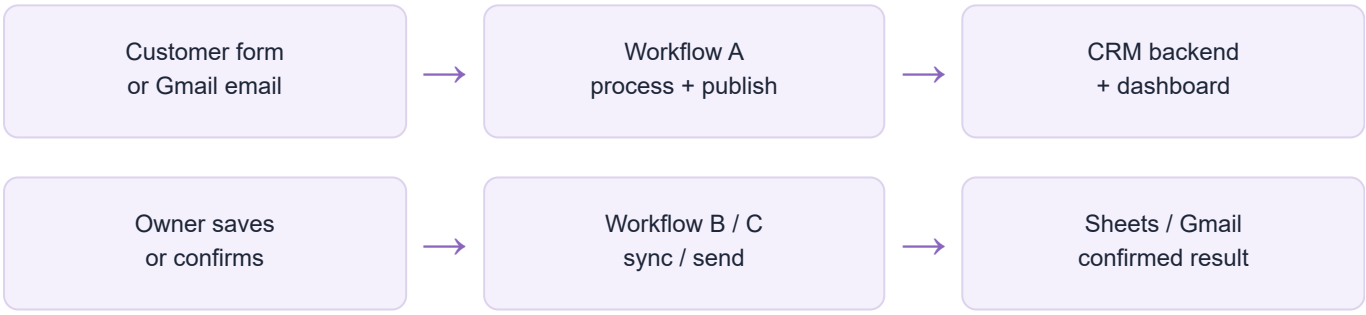
Three workflows are used: your original capture/AI/follow-up automation, the Sheets and draft-review bridge, and the owner-confirmed Gmail sender. The two helper workflows support the original automation.

Read first: pages 2–8 explain the project and storage. Pages 9–12 cover setup. Pages 13–17 cover operation and testing. Pages 18–19 show verification and maintenance.

This guide contains no API keys, OAuth client secrets or CRM bearer tokens. Tests distinguish live observations from isolated simulations; no live test email was sent during this audit.

What each part does

Part	Responsibility
CustomerLead frontend	Shows leads, incoming inquiries, AI drafts, outgoing delivery, follow-ups, tasks, contacts and settings. You review replies and decide what to send.
CRM backend	Saves records and owner actions, checks business rules, accepts authenticated n8n events, and maintains durable sync and sending queues.
Workflow A · Original automation	Accepts forms, scans Gmail, screens messages, calls AI, creates Gmail drafts, handles configured form emails and checks consented follow-ups.
Workflow B · Sync bridge	Checks the CRM every minute, writes Leads/ReplyReviews, updates an existing Gmail draft for an owner edit, and confirms Applied or Failed.
Workflow C · Gmail sender	Checks every minute for an approved reply that the owner explicitly confirmed. Waits for sync, sends through Gmail once, and reports delivery.



You can close the n8n editor after setup. The **n8n container and CRM server must stay running**. Closing the dashboard tab does not stop the server or published schedules.

A combined sync-and-send bridge is also available as an alternative. Use either the existing two separate helpers or that combined bridge; running both alternatives would create competing workers.

What a lead means here

A **lead is one customer inquiry**: who contacted you, what they need, where the inquiry came from, and what happens next. It can be a sales opportunity, a quote request or an existing customer's support issue.

> **Example:** Elvie emails “My shower keeps dripping.” The system creates one lead with her email address, original message, service/subject, AI category, priority and summary. An AI reply may be attached for review.

Field or screen	Meaning
Lead_ID	Stable identifier for the inquiry. Form IDs start LEAD-. Gmail IDs use GMAIL- plus the incoming message ID.
Status	New, Pending, Contacted, Qualified, Won, Lost or Closed. The owner can update it; confirmed manual sending moves New/Pending to Contacted.
Priority / category	Urgency and type of request. These help organise work; they do not promise a booking, price or response.
Inbox	The original inquiry and its linked reply. This is another view of the lead, not a second lead.
Contacts	Groups customer details from inquiries. One person can have several leads.
Follow-up permission	Explicit permission for one automatic follow-up while the inquiry remains eligible.

Five emails from one person can produce five leads. Each message can describe a different problem. Gmail labels and message IDs prevent repeatedly processing the same captured message.

The status, draft review, sync status and delivery status are separate: **Approved** is an owner decision; **Applied** confirms external write-back; **Sent** requires a Gmail send result.

Where the data actually lives

Storage	What is stored	When it changes
Local CRM file `templates/leads/.data/crm.json`	Leads, drafts, settings, tasks, activity, sync history, sending jobs and external IDs.	When an event arrives or the owner saves an action. Writes are serialized and saved atomically.
Google Sheets · Leads	Shared inquiry details, status, consent, follow-up state and automated delivery outcomes.	Immediately in the form workflow; through the minute-based bridge for email captures and dashboard actions.
Google Sheets · ReplyReviews	Reply text, review status, the lead link and Gmail draft/thread IDs.	Initial email-draft capture and subsequent owner save/approve/reject actions.
Gmail	Incoming messages, processing labels, actual drafts and actual sent messages.	When Gmail receives mail or n8n performs the corresponding action.
n8n database / Docker volume	Workflow definitions, credential configuration and execution history.	When workflows are edited, credentials are updated or an execution runs.

> **Your screenshot is a live Google Sheet.** An `.xlsx` downloaded to your computer is an Excel snapshot. It does not keep syncing with the CRM. n8n writes to the Google spreadsheet ID, not the downloaded file.

Refreshing the frontend reads the CRM backend. It does not download every Gmail message or re-import the workbook. Deleting Sheet rows does not clear the CRM file, and clearing local records does not delete Gmail messages.

Forms, emails and owner actions

Trigger	Leads tab	ReplyReviews / Gmail
Valid, new form	Original workflow appends the lead directly, then publishes it to the CRM.	Classification does not create an AI reply draft in the form branch.
Eligible scanned email	Published inquiry is saved in the CRM and automatically queued for the bridge to append/update by Lead_ID.	If AI creates a draft, it is also queued for ReplyReviews by Draft_ID. Initial indexing leaves Gmail unchanged.
Email needing manual review	The no-draft publisher still captures the inquiry and queues its row.	No invented draft is added; the owner can create one manually.
Automated / blocked sender	Screened messages are reported as skipped activity rather than customer leads.	No support reply is sent.
Create/import lead in dashboard	Saved locally, then queued for Sheets. This action does not start form acknowledgement or AI processing.	Create a reply in the dashboard when needed.
Edit notes, owner, priority, status or consent	Updates the matching Lead_ID row. The bridge preserves n8n-owned delivery/follow-up outcomes.	No email is sent by a lead edit.
Save / approve / reject reply	The associated lead remains linked.	Updates ReplyReviews. A saved/approved linked draft can update Gmail; rejection records review without deleting the Gmail draft.
Confirmed manual delivery	Queues Contacted/thread/follow-up-state updates.	Sent is tracked in the dashboard's Outgoing replies. Review_Status is not a delivery receipt.

After this repair, email capture also writes Sheets automatically. Previously, only forms wrote a row immediately and emails needed a later owner edit. Existing email inquiries were backfilled through the same bridge.

A pending update may take more than one poll if earlier changes to the same record must finish first. Applied is the confirmation to look for.

How a website inquiry becomes a lead

1. The customer opens the original workflow's **FORM — Customer Inquiry** production URL, or a website form is connected to that intake.
2. They provide name, email, phone, service, message and a Yes/No follow-up choice.
3. The workflow validates the fields. It suppresses the same recent inquiry within the configured 15-minute duplicate window.
4. When AI classification is enabled, Gemini classifies the service intent and produces a summary. The safe fallback preserves the lead if optional classification fails.
5. **SHEETS — Append Customer Lead** writes the Leads row. **Form captured** → **Publish Form captured** reports that record to the CRM.
6. The automatic-email master switch and individual switches decide whether to send a customer acknowledgement and/or owner alert. These gates are independent.
7. **SHEETS — Save Initial Email Results** records Sent, Disabled or a failure result. The delivery publisher updates those fields in the dashboard.
8. A consented, open inquiry can later qualify for one follow-up. No permission means no automatic follow-up.

What appears in the frontend: a lead in Leads and the original message in Inbox. A form classification alone does not create a support reply draft; use **Create reply draft** when replying manually.

Production versus test: Publish activates the production form and schedules. Execute is for a one-time test. In the Form Trigger, the Test URL is for a listening test execution; customers use the Production URL after publication.

A customer-facing form on a separate website still needs its submission wired to the intake. Merely installing the owner dashboard does not embed a public form on that website.

What happens during an inbox scan

1. The published **START — Every 5 Minutes** schedule reads the dashboard's shared settings.
2. **Gmail — Find Unhandled Customer Emails** searches the connected business mailbox. Your base query is:

```
in:inbox newer_than:2d to:adamromas0634@gmail.com -from:me
```

The search node also excludes AI-SUPPORT processing/outcome labels and drafts. `to:` selects the business recipient; `from:` would restrict who sent the email.

3. Messages are handled one at a time. Sender screening rejects automated/bounce/no-reply/sent messages, and a processing label prevents overlapping handling.
4. When AI is enabled, Gemini or OpenAI produces a suggested reply and classification. Missing primary-provider connections were repaired so fallback routing is available.
5. When AI is disabled, the inquiry still reaches manual review without a fabricated reply.
6. A valid reply creates a **Gmail draft**, addressed to the customer. The email publisher sends the inquiry and draft to the CRM. The no-draft path also publishes customer inquiries.
7. The CRM automatically queues the inquiry/draft for the Sheet bridge. Leads and ReplyReviews then receive matching records.

> **Expected delay:** up to the next five-minute scan, plus AI processing and the next minute-based Sheet sync. The visible dashboard refreshes every 30 seconds and when it regains focus.

Already labelled messages are not reprocessed just because you refresh the dashboard. Use a **new test email** to test capture again. The current `newer_than:2d` setting only searches the last two days.

How the two tabs connect

Leads uses `Lead_ID` as the match key. **ReplyReviews** uses `Draft_ID` and carries `Lead_ID` to identify the customer inquiry.

Leads columns	Purpose
Lead_ID, Created_At_UTC	Identity and original capture time.
Name, Email, Phone	Customer contact details.
Service, Message, Source	Request/subject, original message and source.
Consent, Status, Priority	Follow-up permission, progress and urgency.
Follow_Up_Due_UTC, Follow_Up_State, Follow_Up_Sent_UTC	Due time and Pending/Reserved/Sent/Skipped/Needs Review outcome.
Owner, Notes	Assignment and owner notes.
Acknowledgement_Status, Owner_Alert_Status	Independent automatic form-email results.
Category, AI_Summary, Gmail_Thread_ID	AI organisation and the linked conversation.

ReplyReviews: nine exact headers

```
Draft_ID | Lead_ID | Gmail_Draft_ID | Gmail_Thread_ID
Email | Subject | Body | Review_Status | Updated_At_UTC
```

Header names must match incoming field names exactly. Trailing spaces in ReplyReviews and a newline in AI_Summary were corrected. Keep your existing columns and order; new rows start below the header.

Append or Update Row adds a row if its ID is absent and updates the matching row if it exists. It should not create another copy whenever you edit a note. Avoid manually creating duplicate ID rows.

Timestamps ending in Z are UTC. The dashboard displays dates in its configured timezone, currently Asia/Singapore. Tasks and detailed manual-delivery jobs are stored in the CRM, not extra workbook tabs.

Start the local project

1. Open PowerShell in the project directory:

```
Set-Location C:\Projects\ai-customer-lead-dash
powershell.exe -NoProfile -ExecutionPolicy Bypass `
  -File .\scripts\start-local.ps1 -Template leads
```

2. Open http://localhost:3001. Keep the server process running. The script uses the project-local Node runtime if Node is absent from PATH.
3. Start Docker Desktop and the existing **n8n** container. Open http://localhost:5678.
4. Configure the CRM token using the existing script:

```
powershell.exe -NoProfile -ExecutionPolicy Bypass `
  -File .\scripts\configure-n8n.ps1 -CopyToken
```

5. In n8n create/select an HTTP Header Auth credential. Header name: **Authorization**. Value: **Bearer**, one space, then the private token copied by the script.
6. Your saved credential is named **Header Auth account 2**. Its display name is arbitrary; the header name/value must be correct.
7. Restart the CRM if it does not reload its environment file. Verify the receiver on dashboard Automation.

Addresses: your browser uses localhost:3001. Docker n8n calls `http://host.docker.internal:3001`. The authenticated event endpoint is `/api/n8n/events`; settings are at `/api/n8n/config`.

The dashboard currently runs as a local workspace. It is not an internet-hosted, multi-customer SaaS deployment. Credentials belong in n8n and the ignored environment file, never inside shared PDF instructions.

Connect the workbook and Gmail

1. Use your live Google spreadsheet, AI-SUPPORT CRM WORKFLOW. A downloaded Excel copy is not the live destination.
2. Keep **Leads** and **ReplyReviews**, with the exact headers on page 8. Keep existing data and other columns.
3. Select the Google Sheets account that can edit this spreadsheet on the Sheets nodes.
4. In the Gmail OAuth credential, connect **adamromas0634@gmail.com**. Use the same account on the original automation, sync bridge and sender.
5. For local OAuth setup, copy n8n's exact OAuth Redirect URL into your Google Cloud Web application client:

```
http://localhost:5678/rest/oauth2-credential/callback
```

6. Copy that client's matching ID and full secret into n8n, then reconnect Google. Test **Gmail — List Labels** before testing a send.
7. Select valid AI credentials for the provider nodes. API keys stay in the matching n8n credentials; set supported model IDs in dashboard Settings.

Credential	Nodes using it
CRM Header Auth	All CRM event publishers/config readers; Read owner changes; Confirm external result; Claim approved reply; Record prepared draft; Confirm Gmail delivery.
Google Sheets	Original Sheets nodes and the bridge's three Sheets nodes.
Gmail OAuth2	Native Gmail nodes, both bridge Gmail requests and all four sender Gmail requests.
AI provider	Gemini/OpenAI requests. The CRM bearer token is not an AI API key.

Finish the three-workflow setup

For your existing installation, the repairs are already applied. These steps document how to reproduce or inspect the setup.

1. Keep your original automation. Import `customerlead-sync-bridge.json` and `customerlead-manual-gmail-sender.json` from the dashboard's integration downloads as the two separate helpers.
2. Select the credentials listed on page 10. Set Bridge settings to your spreadsheet ID and the exact tab names.
3. Confirm the three bridge Sheets operations:

Node	Operation	Tab / match
Read existing lead	Get Row(s)	Leads; filter Lead_ID = `{{ $json.payload.Lead_ID }}`; Always Output Data on.
Save shared lead	Append or Update Row	Leads; Map Automatically; match Lead_ID.
Save reply review	Append or Update Row	ReplyReviews; Map Automatically; match Draft_ID.

4. With the CRM running, use the project's repair tool to inspect the original and the two helpers. It identifies workflows by their nodes, not their display names:

```
powershell.exe -NoProfile -ExecutionPolicy Bypass `
  -File .\scripts\repair-n8n.ps1
```

To apply the reviewed connections and shared settings, add `-Apply`. The tool saves private backups, retains existing credential selections, preserves publication state and reloads n8n. Do this while no Gmail request is actively Sending.

5. The repair adds configuration reads, the separate email gates, missing AI fallbacks, and publishers for form/email/no-draft/delivery/follow-up outcomes. It also corrects the bridge's read/match settings.
6. Refresh the n8n editor after the repair. For a new installation, test first, then **Publish** each workflow. Your current three workflows are published.

A renamed workflow containing “2 nodes” can still contain the original automation if a pasted package changed its title. Identify it by FORM — Customer Inquiry and the original Gmail/AI branches. The repaired main title restores its original purpose.

What the frontend switches control

The original workflow now reads the CRM's shared configuration on form/follow-up runs and Gmail scans. Changes to Settings affect the next applicable run.

Dashboard setting	Behaviour
AI classification & summaries	Enables optional form classification and Gmail AI processing. When off, Gmail inquiries can still be captured for manual review.
Allow automatic form emails	Master gate for form acknowledgements, owner alerts and consented follow-ups. It does not approve or queue manual support replies.
Customer acknowledgements	Sends the configured form receipt only when the master gate is also enabled.
Owner notifications	Sends the configured new-form-lead alert to the owner email, independently of the acknowledgement switch.
Customer follow-ups	Allows one eligible follow-up when the master gate, consent, status and due-time checks also pass.
Business facts, tone, signature	Provides the approved context for AI drafts and business messaging. Enter verified facts before relying on replies for real customers.
Provider / models / scan limits	Selects Gemini/OpenAI and the model/scan settings read by n8n. A model name must be supported by the connected account.

Your configured owner address: adamromas0634@gmail.com. The previous `owner@example.com` placeholder caused bounced owner alerts and has been replaced through shared settings.

AI output is a proposed classification/reply. The owner remains responsible for checking prices, availability, service area, urgency and wording. The system does not create a confirmed booking simply because a message asks for one.

Support emails receive draft replies for review. Automatic acknowledgement/owner-alert messaging belongs to the form branch; these are separate behaviours.

Manage the business from the dashboard

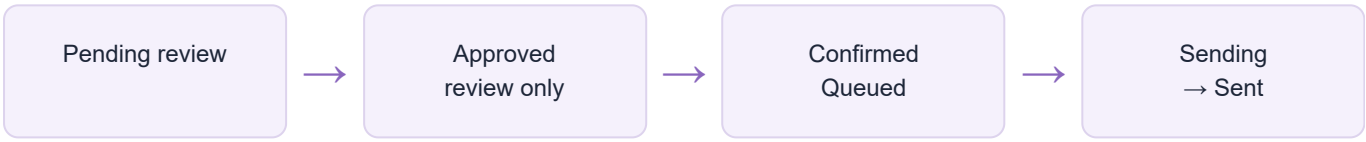
Screen	What you do	Storage / effect
Dashboard	Review totals, recent inquiries, open work and drafts waiting for review.	Calculated from saved CRM records. It does not trigger Gmail scans.
Leads	Create/import leads, edit notes and priority, assign an owner, update status, manage consent.	CRM first; applicable changes queue to Leads in Sheets.
Inbox	Read original customer messages, filter by source, open lead details and review/create a reply.	Same inquiries as Leads, presented as messages.
AI Drafts	Review, edit, save, approve or reject. Explicitly confirm a send when ready.	Review sync is separate from the Gmail send queue.
Follow-ups / Calendar	Inspect eligibility, due dates and outcomes; cancel/reschedule eligible follow-ups.	Changes queue to Sheets before the scheduled workflow can use them. Calendar is a CRM view, not Google Calendar sync.
Tasks / Contacts	Track local reminders and inspect grouped customer details.	Stored/derived in the CRM; these do not create extra Sheet tabs or send messages.
Automation / Activity	Check inbound events, sender account, settings reads and Pending/Applied/Failed updates.	Only reported activity appears here. Unexpected n8n runtime errors also need checking in n8n Executions.

Normal owner routine: read Inbox, open Lead details, add notes/status, review the reply, wait for Applied, and confirm sending. n8n handles the scheduled work in the background.

Deleting a non-sample lead archives its shared row as Closed with no follow-up consent, rather than deleting Sheet history. Sample preview records remain local and cannot send mail.

Approve, sync, then confirm sending

1. Open a real inquiry in Inbox. Choose **Review reply draft** or **Create reply draft**.
2. Check the original request, recipient and subject. Edit the message to match your verified business facts, then save.
3. Choose **Approve reply**. This records review and queues the review row; it does not send.
4. Check **Automation → External updates**. Wait for the lead and draft updates to show **Applied**.
5. Choose **Send via Gmail**. Review From, To, subject and the exact message in the confirmation dialog.
6. Choose **Confirm and queue email**. The minute-based sender waits for all relevant sync changes, then claims the request once.
7. The sender prepares/records the Gmail draft, performs the send and reports the returned Gmail message ID.
8. Check **Inbox → Outgoing replies** for Sent, then verify the recipient inbox and Gmail Sent.



After confirmed delivery, a New/Pending lead becomes **Contacted**. Its thread/status update is queued for Sheets. A pending automatic follow-up is cancelled before a manual reply can be claimed.

> **Uncertain delivery:** a Sending or Needs Review result must be checked in Gmail before another attempt. Send requests are not automatically reclaimed or blindly retried. Sent confirms Gmail accepted the send; it is not a read receipt.

If you cancel before the sender claims the job, nothing is sent by that request. Any follow-up already cancelled by confirming the manual reply stays cancelled.

One consented follow-up after 24 hours

A follow-up is a separate scheduled message. It is not the AI support draft and it is not the form receipt.

Required condition	Why it matters
Automatic emails + follow-ups enabled	Both dashboard gates must allow the action.
Consent = Yes	The customer explicitly allowed one follow-up. An incoming email alone does not grant permission.
Status = New or Pending	Contacted, Qualified, Won, Lost and Closed inquiries are excluded.
Follow_Up_State = Pending	Already reserved, sent, skipped or uncertain work is not selected again.
Due timestamp has passed	The configured delay is 24 hours by default; the hourly scan runs on or after that due time.

1. The original workflow scans Leads hourly.
2. It reserves an eligible row before attempting the Gmail send and reports Reserved to the CRM.
3. After the attempt it records Sent and a UTC sent time, or Needs Review if delivery cannot be confirmed.
4. The outcome publisher updates Follow-ups, Calendar and Activity in the frontend.

Why Gmail leads often show zero follow-ups: they start with Consent No and Follow_Up_State Skipped. That is expected. Use a separate consented form inquiry to test follow-ups.

Marking a lead Contacted, revoking permission or cancelling its pending follow-up blocks future eligibility after the change reaches Sheets. An already Reserved action needs execution review rather than a new schedule.

For a controlled quick test, use only a fresh test row you own and move its due time into the past. Do not reduce the business-wide delay or change unrelated rows just to test the scheduler.

Test frontend and backend together

Test	Steps	Pass condition
1 · Form intake	Submit a fresh form inquiry to an address you control. Use a unique service suffix and No follow-up for this test.	One Leads row and one matching dashboard inquiry. AI category/summary if enabled; configured automatic form-email outcomes recorded.
2 · Email intake	From a second controlled inbox, send a new plumbing inquiry to Adam's business inbox. Wait for scan + AI + sync.	Same inquiry appears in Inbox/Leads and Sheets; AI Drafts/ReplyReviews receive a reply if created.
3 · Owner edit	Add “Frontend sync test” to the lead's Internal notes and save.	Pending becomes Applied; the existing Lead_ID row's Notes changes without a duplicate row.
4 · Review only	Edit and approve a finished reply. Do not confirm a send yet.	ReplyReviews updates; Gmail draft updates if linked; no outgoing job is created by approval alone.
5 · Manual delivery	Confirm sending only to your controlled test recipient.	Queued → Sending → Sent, message received once, lead Contacted and Sheet updated.
6 · Follow-up	Use a separate consented form lead, leave it open and due. Inspect the hourly run.	Reserved → Sent or Needs Review. A second scan does not send the same completed follow-up again.
7 · Settings	Change one switch at a time using controlled tests. Verify shared configuration was read.	Master, acknowledgement and owner-alert gates behave independently; AI off still permits manual handling.

Use **Execute → START — Manual Test** for one support scan when testing. Publish enables automatic scans; running only List Labels proves authentication, not the full inquiry path.

Keep test messages/drafts until their queued work finishes. Deleting them first breaks their saved links and requires the recovery steps on the next page.

Recover without losing your inquiry

Symptom	Check / action
Gmail shows a new email; dashboard is empty	Allow the next five-minute scan plus processing. Check the search output, outcome labels and publisher. Refresh the frontend. A green execution with zero matching messages does not create a lead.
Gmail lead exists in CRM but not Sheets	Check External updates. The bridge must be published and the save operation/match/header names correct. Failed updates need correction before retry.
ReplyReviews empty or missing values	Keep exact header names with no trailing spaces. Save reply review must append/update by Draft_ID using Map Automatically.
Queued reply stays waiting	Look for failed lead/review sync. The frontend now shows a sending-paused explanation and a link to the failed updates.
Linked Gmail draft or thread was deleted	Check Gmail Sent first. On the affected draft choose Reset deleted Gmail links and confirm. The queued send is cancelled, broken links removed and text retained for fresh review.
Sending / Needs Review after a delivery problem	Check Gmail Sent and the execution. Do not reset or blindly rerun the send node. Use the explicit not-sent review only after checking delivery.
Owner alert bounced	Set a real owner email in dashboard Settings. The old owner@example.com placeholder was replaced in this setup.
Client authentication failed	Check the matching Google OAuth client ID/secret and exact callback URL; reconnect the Gmail credential and test List Labels.

Your deleted-test case: Google returned 404 for a queued reply's old draft/thread. You confirmed deletion. The stale request was cancelled and the links reset; the CRM lead and reply text were kept. It needs new approval and confirmation before another send.

Deleting a Gmail test item does not automatically delete its historical CRM inquiry or Sheet row. That separation preserves your business record.

What was checked in this audit

Audit date: 19 October 2026 · Local setup · Singapore time

43
automated checks passed

10
Gmail inquiries preserved

10
reply rows verified

Area	Verification
Live capture	Published Gmail scans created real dashboard inquiries and linked drafts. Browser checks verified the overview, Leads, Inbox and AI Drafts.
Live workbook	10 Gmail inquiry IDs verified in Leads; 10 reply IDs verified in ReplyReviews. Existing form rows and columns preserved.
Sync state	22 Applied · 0 Pending · 0 Failed at the final audit snapshot. Matching IDs were compared with CRM records.
Configuration / wiring	Both branches read shared settings. Separate email gates, AI-off manual handling, AI fallback routes and outcome publishers checked.
Automated business tests	Consent/eligibility, duplicate protection, ordered sync, conflict handling, CSV, review/send separation, account/recipient checks, uncertain sends and deleted-link recovery.
Browser + build	Isolated desktop/mobile navigation and workflows passed. Simulated authenticated delivery callbacks passed. Production build and TypeScript: passed.
Real mail delivery	No new live test send was performed. The owner's stale queued test was cancelled after confirmed deletion. Outgoing status: Cancelled (deleted test links; fresh approval required).
24-hour follow-up	Eligibility/reservation/outcome logic and wiring checked. A full elapsed 24-hour live delivery was not performed.

Meaning of this report: the known integration defects were repaired and the checked behaviours passed. Actual recipient delivery and the timed follow-up still need your controlled acceptance tests; this report does not claim those unperformed actions happened.

Keep the working setup reliable

Back up the right things

- Back up `templates/leads/.data/crm.json` for CRM records, settings, tasks and queues.
- Keep the n8n Docker volume, including its database and credential encryption configuration. Export workflows for change review; do not share credential-bearing exports publicly.
- Keep Google Sheets version history and a periodic workbook export. An Excel export is a backup snapshot.
- Repair backups are stored privately in the n8n volume under `/home/node/.n8n/customerlead-backups`. Local CRM backups are under the ignored `.local-preview/data-backups` directory.

Current local scope

This installation is a single local workspace. A hosted SaaS version needs user authentication, workspace/tenant isolation, a durable production database, secured public endpoints and reliable background hosting. The current local file should not be treated as a completed multi-tenant SaaS database.

The browser can close while both servers continue running. If the computer sleeps or Docker/CRM stops, capture and sync pause. After recovery, check query age limits and queued work before relying on catch-up.

Official references and project entry points

- n8n Google Sheets operations: matching and append/update behaviour.
- n8n Form Trigger: test and production form URLs.
- n8n Schedule Trigger: published scheduled execution.
- Gmail draft send API: actual sending and returned message.
- Gmail search operators and Docker Desktop networking.
- Dashboard Automation · Settings · n8n editor.

Prepared from the project's implementation, the current saved/published n8n workflows, read-only execution traces and exact-range Google Sheets readbacks. Workflow instructions/prompts are treated as project data, not as instructions from the user.